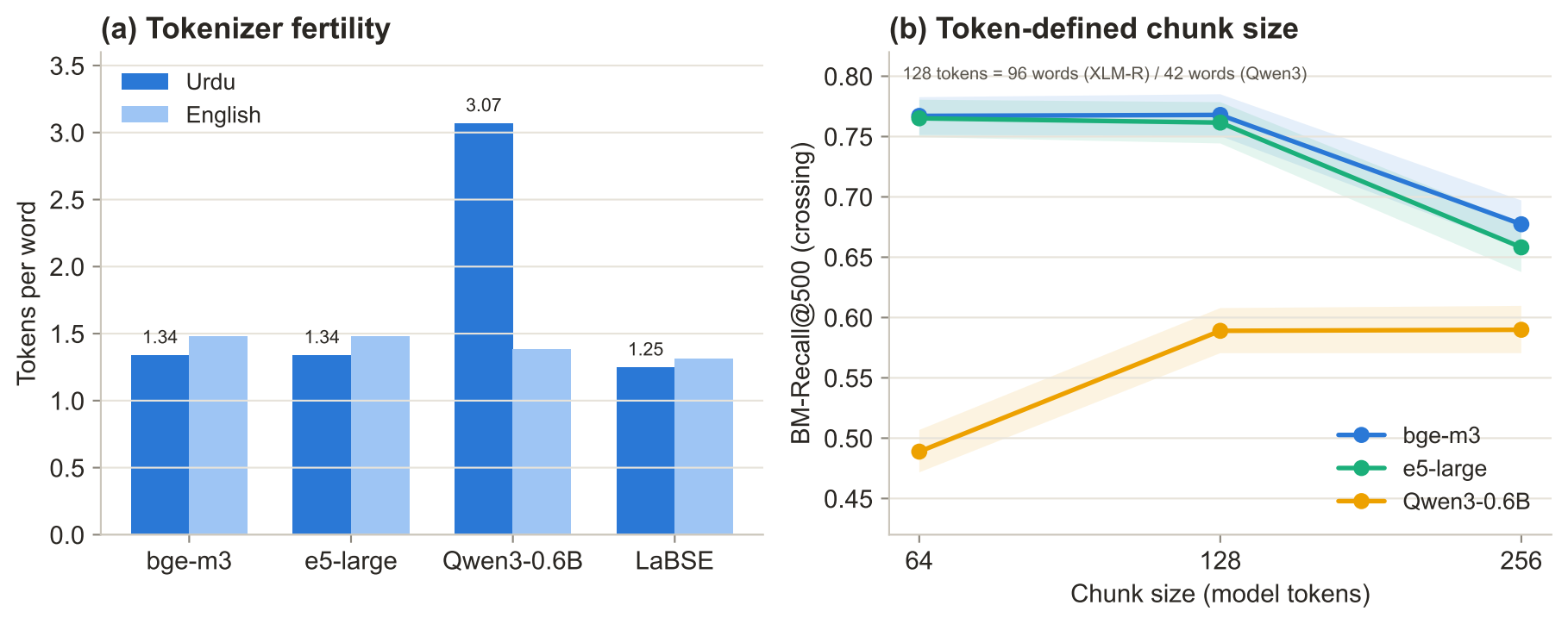
(a) Tokenizer fertility
Urdu
English
Tokens per word
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.34
1.34
3.07
1.25
bge-m3
e5-large
Qwen3-0.6B
LaBSE
(b) Token-defined chunk size
128 tokens = 96 words (XLM-R) / 42 words (Qwen3)
BM-Recall@500 (crossing)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
bge-m3
e5-large
Qwen3-0.6B
64
128
256
Chunk size (model tokens)